

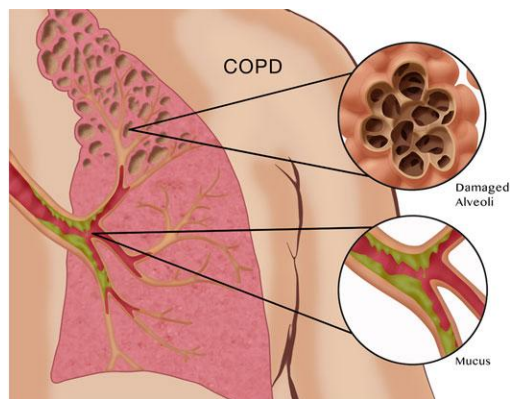
The Mermaid Theory: Pulmonary Exchange via Circulating Oxygen-Rich Inert Fluids

AI tools were used in drafting and language refinement of this paper; sources were verified and cited appropriately by the author.

Author: Penultimate pharmacy student Abbas J.K. Dherish

1. Introduction

Chronic obstructive pulmonary disease (COPD) is characterized by airflow limitation, chronic inflammation, and mucus hypersecretion that together led to reduced gas exchange and progressive respiratory failure. Severe COPD patients may eventually lose critical alveolar function, and conventional therapies (such as noninvasive ventilation and long-term oxygen therapy) are often used in an attempt to augment oxygenation and reduce exacerbations (1.1). A novel therapeutic idea is to use an oxygen-rich perfluorocarbon liquid (oxyfluor) in a liquid ventilation modality to “support” the impaired lungs, **based on the physical principle that liquids can dissolve and transport more**



oxygen than gas. Proponents theorize that such a fluid may not only supply oxygen more effectively in a lung with diminished alveolar surface area but also, through its viscosity and kinetic properties, mechanically mobilize mucus from

obstructed airways. **When combined with mucolytics that reduce mucus viscosity and bronchodilators that widen the airways**, the approach might improve both oxygen delivery and mucus clearance in severely compromised COPD lungs (2.1).

2. What are Perfluorocarbons (PFCs)?

Perfluorochemicals (PFCs) are a group of synthetic chemicals that have been used in many consumer products. The structure of these chemicals (Figure 2) makes them very stable, hydrophobic (water-repelling), and oleophobic (oil-repelling).

Simply PFCs are synthetic compounds capable of dissolving large amounts of oxygen and carbon dioxide. They function by physically dissolving oxygen in their

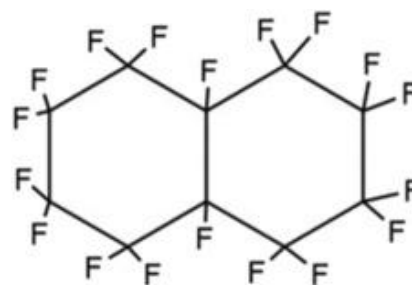


Figure 2

structure and releasing it at the site of need. **And in comparison**, to Hemoglobin-Based Oxygen Carriers (HBOCs), HBOCs use modified hemoglobin molecules to transport oxygen throughout the body. These molecules can be derived from human, bovine, or recombinant sources.

Both HBOCs and PFCs represent significant advancements in the development of blood substitutes, each with unique mechanisms, advantages, and challenges. HBOCs, with their hemoglobin-based oxygen delivery, offer a promising alternative to blood transfusions, especially in emergency and surgical settings.

However, their potential side effects and compatibility issues must be managed.

PFCs, while less efficient in oxygen delivery, provide a stable and non-biological option, beneficial in specific medical scenarios and regions with limited blood supply.

Oxyfluor™

Oxyfluor is one of several ‘second generation’ perfluorocarbon emulsions to be summarized in this volume. Like Oxygent™, Oxyfluor has a significantly higher oxygen capacity than Fluosol-DA. (Table 1)

Study	Specie(s)	Result(s)
CV pharmacology	Dog	Does not affect cardiovascular function in the dog.
Mutagenicity	Salmonella typhimurium	Negative (-)

Acute toxicity	Baboon	No adverse pulmonary events.
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3. Mermaid Technique - تقنية الحورية

3.1: Theoretical Rationale and Background

Liquid ventilation involves instilling a biocompatible, oxygen-dissolving liquid—typically a perfluorocarbon—into the lungs, where it serves as a medium for gas exchange while simultaneously reducing surface tension and facilitating alveolar recruitment (2.2). Perfluorocarbon liquids have shown high solubility for both oxygen and carbon dioxide; thus, they can theoretically deliver far more oxygen per unit volume than conventional air or oxygen mixtures. This is particularly attractive in the setting of severely depressed or “functional” lungs, as is often seen in advanced COPD, where alveolar destruction and airway obstruction impede the efficient delivery of oxygen from air (2.1).

Furthermore, the physical properties of such liquids—characterized by low surface tension and high density—could assist in mobilizing mucus by exerting a “sweeping” action on mucus deposits along alveolar and airway surfaces. Experimental studies in acute lung injury have shown that the kinetic forces of liquid ventilation might mechanically displace mucus, thereby facilitating its removal and improving ventilation–perfusion matching (3.1, 4.1).

While these studies were conducted primarily in the context of [acute respiratory distress syndrome \(ARDS\)](#) and [acute lung injury \(ALI\)](#), they provide a theoretical foundation for extrapolating some potential benefits to severe COPD, *even though no clinical trials have specifically confirmed such effects in the COPD*. **Importantly**, combining the liquid ventilation approach with pharmacologic therapies such as mucolytics (which reduce mucus viscosity) and bronchodilators (which relax airway smooth muscle and dilate airways) may further promote the clearance of mucus plugs. The improved clearance combined with enhanced oxygen delivery could maintain or even improve gas exchange in an otherwise severely compromised lung (2.2). **Thus**, the conceptual framework relies on both the **physicochemical** benefits of oxyfluor and the **synergistic** effects of conventional pharmacologic agents.

3.2: Potential Physiological Benefits

a) Enhanced Oxygen Delivery

Fundamental to the concept of using perfluorocarbon liquids is their high oxygen-carrying capacity. Unlike air, which contains only 21% oxygen by volume, perfluorocarbon liquids can dissolve substantially higher concentrations of oxygen due to their unique chemical structure. In studies focusing on partial liquid ventilation in ALI/ARDS, these liquids have been demonstrated to improve oxygenation by delivering oxygen directly to alveolar spaces, thereby sustaining gas exchange even when traditional air ventilation is less effective (2.1, 2.2).

Extrapolating from these results, it is

conceivable that in severe COPD—where damaged alveoli and extensive mucus plugging reduce the effective alveolar surface area—the instillation of an oxygen-rich liquid could provide supplemental oxygen directly to regions of the lung that are otherwise poorly ventilated.

b) Improved Mucus Clearance

The challenges of mucus hypersecretion and impaired clearance are central to COPD pathophysiology. The dense, viscous mucus that accumulates in diseased airways contributes to airflow obstruction and increases the risk of infections. The physical properties of perfluorocarbon liquids, notably their low surface tension, may reduce mucus adhesion and allow the liquid to “wash” or mobilize secretions from the airway walls. Experimental evidence from animal studies (See table 1) suggests that instilling liquid ventilation agents can alter the rheologic properties of mucus, making it less viscous and more amenable to clearance (3.1). In addition, studies using high-frequency oscillatory ventilation combined with partial liquid ventilation have reported improved lung mechanics and reduced indicators of lung injury, partially attributable to better mucus displacement (4.2). In theory, if a perfluorocarbon liquid such as oxyfluor were applied in COPD patients, it might help “suck” or mobilize thick mucus from the airways, particularly when used in conjunction with mucolytic agents that chemically break down mucus structure (Clinical Trial Search: mucolytics AND bronchodilators AND COPD AND liquid ventilation).

c) Synergistic Effect with Adjunct Therapies

The proposed strategy is not meant to replace conventional therapies but to serve as an adjunct, particularly in situations where standard air/oxygen ventilation is no longer sufficient to support gas exchange. In severe COPD patients, conventional oxygen therapy or noninvasive ventilation has demonstrated benefits but may ultimately be limited by the extent of airway obstruction and parenchymal damage (1.1). By temporarily “recruiting” additional alveolar units and facilitating mucus clearance, liquid ventilation with oxyfluor could complement the use of conventional mucolytics and bronchodilators. Bronchodilators relax bronchial smooth muscle, while mucolytics reduce mucus viscosity; together, they create an environment in which the fluid—if administered properly—can cover airway surfaces more uniformly and permit improved gas exchange (2.1, 3.1). Although preclinical and clinical evidence has so far been restricted to settings of ALI and ARDS, the underlying mechanisms suggest that a similar combination might benefit COPD patients with severe respiratory compromise.

3.3: Limitations and Safety Concerns

- a) Lack of Direct Clinical Evidence in COPD.
- b) Adverse Events Reported in Acute Lung Injury Studies.
- c) Technical and Procedural Challenges.
- d) Patient Tolerability and Experience.

4. Discussion

In summary, the theoretical proposition of using oxyfluor liquid ventilation in severe COPD hinges on two central physiological promises: 1) the capacity to enhance oxygen delivery by using a high-oxygen-carrying liquid, and 2) the ability to mechanically mobilize mucus via the liquid’s physical properties. The experimental evidence drawn from studies in ALI/ARDS supports the notion that perfluorocarbon liquids can improve gas exchange, *at least transiently*, and possess unique properties that may influence mucus clearance (2.1, 4.1). The theoretical synergy with mucolytics and bronchodilators is compelling. Mucolytic agents—by reducing mucus viscosity—and bronchodilators—by increasing airway caliber—could serve to enhance the distribution and efficacy of the liquid medium within the lungs. **In principle**, the combined therapy could overcome the dual challenges of inefficient oxygen delivery and mucus obstruction that worsen as COPD progresses (2.1).

5. Further Reading

- i. [Chapter 27 - Rational Development of Oxyfluor™.](#)
- ii. [Acute Respiratory Distress Syndrome \(ARDS\) and Acute Lung Injury \(ALI\) Revisited by Dr.Dhaher JS Al-habbo.](#)
- iii. [Biomonitoring | Perfluorochemicals \(PFCs\).](#)